Arvant Motors

Technical Specification

RFQ AM-2026-0417 - Volt-7 electric platform

Issued to supplier: Acme Components

Issue date: 2026-09-14

Response due: 2026-10-30

Buyer contact: Ingrid Solberg, Senior Commodity Buyer, sourcing@arvant-motors.example

This document is a fictional sample created for software testing. All names, figures and references are invented. Standards are cited by name and number only.

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Each numbered clause states one requirement. The Supplier answers every clause in the questionnaire and in the response letter using one of five statuses.

Section 1 - Scope and definitions

This section defines the scope of the request and the terms used in this document.

1.1 This specification applies to all parts for the Volt-7 electric platform that Acme Components is invited to quote under RFQ AM-2026-0417, covering the six part families listed in section 2.

1.2 In this document Buyer means Arvant Motors, Supplier means Acme Components, and Part means any item supplied to a Buyer drawing, specification or purchase order issued under this RFQ.

1.3 Where documents conflict, the Buyer drawing prevails over this specification, and this specification prevails over any referenced Buyer standard.

1.4 The Supplier quotation shall remain valid for 180 calendar days from the response due date and shall be submitted in English.

1.5 Start of production is planned for September 2027; the Supplier shall confirm that tooling and capacity can support this date.

Section 2 - Technical requirements

Numeric values in this section are minimum performance levels unless stated otherwise.
Sub-sections follow the six part families in scope.

Sub-section HSG - Aluminium die-cast housings

2.1 Aluminium die-cast housings for the EV motor and inverter shall operate continuously from -40 C to +150 C without loss of function.

2.2 Housing castings shall achieve a tensile strength of at least 240 MPa measured on cast-to-shape test bars.

2.3 Housings shall withstand 720 h of neutral salt spray with no corrosion penetrating the coating.

2.4 Sealed housings shall meet ingress protection IP6K9K in the fully assembled condition.

2.5 The sealing face of each housing shall be flat within 0.15 mm.

2.6 Housings shall pass the thermal shock sequence defined in the Buyer test plan, which will be issued after nomination.

Sub-section FST - Forged steel fasteners class 10.9

2.7 Class 10.9 forged steel fasteners shall have a tensile strength of at least 1040 MPa.

2.8 Fasteners shall carry a zinc-nickel coating applied to the Buyer coating specification.

2.9 Coated fasteners shall withstand 480 h of salt spray to red rust.

2.10 Fasteners shall retain their rated properties from -40 C to +200 C.

Sub-section SNS - PA66-GF30 sensor housings

2.11 PA66-GF30 sensor housings shall operate from -40 C to +125 C.

2.12 Sensor housings shall meet ingress protection IP67.

2.13 Metallic inserts and coated surfaces of sensor housings shall withstand 240 h of salt spray.

Sub-section CON - Terminal and connector assemblies

2.14 Terminal and connector assemblies shall operate from -40 C to +125 C.

2.15 Connector assemblies shall meet ingress protection IP67 when mated.

2.16 Connector assemblies shall carry a rated current of 30 A continuously.

2.17 Connector assemblies shall withstand 10000 cycles of mating and unmating without loss of contact performance.

2.18 High-current connector variants for the traction inverter shall be rated at 40 A in addition to the standard 30 A variant.

Sub-section BRK - Ductile iron brake caliper brackets

2.19 Ductile iron brake caliper brackets shall achieve a tensile strength of at least 500 MPa.

2.20 Brackets shall withstand 480 h of salt spray.

2.21 Brackets shall operate from -40 C to +180 C.

2.22 The Supplier shall provide fatigue life data for brackets to the Buyer load spectrum.

Sub-section HRN - Wiring harness sub-assemblies

2.23 Wiring harness sub-assemblies shall operate from -40 C to +105 C.

2.24 Harness connectors shall meet ingress protection IP67 at the connector.

2.25 Every harness sub-assembly shall pass a 100 percent end-of-line electrical test before shipment.

Section 3 - Materials and environmental

This section covers material declarations and environmental obligations.

3.1 The Supplier shall submit a material data sheet through the IMDS system for every Part before first sample submission.

3.2 All Parts shall comply with the REACH and RoHS regulations and the Supplier shall provide a written declaration on request.

3.3 The Supplier shall maintain an environmental management system certified to ISO 14001 at every producing site.

3.4 Parts shall comply with the Buyer restricted substances list, which is revised twice a year; the applicable revision will be confirmed after nomination.

3.5 Packaging materials shall be recyclable and free of single-use plastics where technically feasible.

Section 4 - Quality

This section covers quality systems, certifications and quality performance.

4.1 The Supplier shall hold a valid IATF 16949 certificate covering every site that produces Parts under this RFQ.

4.2 The Supplier shall hold a valid ISO 9001 certificate for all producing sites.

4.3 The Supplier shall hold a valid ISO 45001 certificate for occupational health and safety at all producing sites.

4.4 The Supplier shall hold a valid TISAX assessment label at assessment level AL2 for information security.

4.5 Production part approval shall be submitted at PPAP level 3 unless the Buyer agrees a lower level for carry-over parts.

4.6 Delivered quality shall be below 50 PPM, measured monthly per part number.

4.7 The Supplier shall respond to a quality complaint with a complete 8D report within 10 working days.

4.8 The Supplier shall submit a control plan and a process FMEA for each Part with the PPAP package.

Section 5 - Logistics

This section covers lead times, delivery terms and data exchange.

5.1 The tooling lead time from tooling order to first sample shall not exceed 16 weeks.

5.2 The series lead time from release to delivery shall not exceed 6 weeks.

5.3 The Supplier shall deliver under Incoterms 2020 FCA at the named Supplier site.

5.4 The Supplier shall also quote delivery under Incoterms 2020 DAP at the Buyer assembly plant.

5.5 The Supplier shall exchange delivery schedules and despatch advices with the Buyer through EDIFACT EDI messages.

5.6 In addition to EDIFACT messages, the Supplier shall send the advance shipping notice through the Buyer web API.

5.7 The Supplier shall submit a packaging proposal for Buyer approval before the first series delivery.

5.8 The Supplier shall hold a safety stock of finished Parts equal to one week of average demand.

Section 6 - Commercial and legal

This section covers commercial and legal conditions. The commercial terms document restates them.

6.1 The Supplier accepts liquidated damages for late delivery of up to 5 percent of the value of the affected purchase order.

6.2 The Supplier shall grant an annual price reduction of 2 percent for 3 years from start of production.

6.3 Payment shall be made 75 days net from the date of a correct invoice.

6.4 The Supplier warrants Parts against defects for 36 months or 100000 km, whichever comes first.

6.5 Supplier liability under this RFQ is limited to direct damages caused by a defective Part.

6.6 Tooling paid for by the Buyer remains the property of the Buyer and shall be marked as such.

6.7 Each party retains ownership of its background intellectual property; no licence is granted except as needed to supply Parts.

6.8 The Supplier shall accept the Buyer Supplier Code of Conduct in its current version.

6.9 The Buyer may audit the Supplier and its sub-suppliers on 10 working days notice.

Annex A - Document register

The documents below are referenced by this specification and are available on the Buyer supplier portal after the supplier registers.

Document	Title	Revision
DOC-4425	Part drawings package	R3
DOC-7075	Quality manual requirements	R4
DOC-1331	Packaging guideline	R2
DOC-7346	Logistics manual	R1
DOC-9872	Supplier code of conduct	R5
DOC-2206	Test plan index	R4

Annex B - Submission instructions

Step 1: Submit the response through the Buyer portal before the due date shown on the cover page.

Step 2: Provide one response letter, the completed questionnaire workbook and the commercial terms mark-up.

Step 3: Answer every clause using exactly one status: Comply, Partial, Deviate, Exception or Clarify.

Step 4: Where a status is not Comply, state the reason and, where possible, the alternative offered.

Step 5: Questions about this RFQ must be sent in writing to the Buyer contact on the cover page.

Step 6: Answers to questions are shared with all invited suppliers without naming the asker.

Step 7: Reference your submission with the supplier number SUP-59235.

Annex C - Evaluation criteria

Offers are scored on the weighted criteria below after the technical clarification round.

Criterion	Weight
Technical compliance	35 percent
Quality systems	20 percent
Commercial terms	30 percent
Logistics and delivery	15 percent

Annex D - Revision history

Revision	Date	Change
R0	2026-09-14	First issue

Annex E - Glossary

8D - Eight-discipline problem solving report format.

EDIFACT - Electronic data interchange message standard family.

IMDS - Material data system used to declare vehicle materials.

PPAP - Production part approval process, submitted at levels 1 to 5.

PPM - Defective parts per million delivered.

TISAX - Information security assessment and exchange mechanism.